



CATOLICA
LISBON
B/SINESS & ECONOMICS

RESEARCH PAPER

Did COVID-19 change people's behavior when they left home?

Abstract

This paper analyzes how mobility behavior of people shifted throughout the COVID-19 pandemic based on shifting new case rates and policy interventions. Following the evidence of Chetty et al. (2023), two city-level regressions for alternative weeks were estimated: one that forecasts the rate of newly reported cases of COVID-19 based on historic case levels, deaths, vaccination levels, and state-level policy interventions, and the other that estimates to what degree randomness in the new rate of cases influence time spent away from home. The result of the first regression reveals that early pandemic interventions, such as school and business closures, played a significant role in lowering new case rates. Vaccination rates then took over as the primary driver following the easing of lockdowns. The second model determines that unexpected increases in new case rates led to reductions in mobility, revealing that individuals respond more to unforeseen changes as opposed to total case rates. These findings signify the uncertainty of affecting behavioral response to health crises.

Silvia Busquets Xapellí
156624031

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Introduction

The COVID-19 pandemic has influenced human behavior, resulting in changes in daily routines, social interactions, and mobility. The crisis has affected how people work, shop, socialize, and travel. Governments worldwide reacted to the spread of the virus by implementing various interventions, such as lockdowns, remote work mandates, and social distancing measures. Although these interventions mitigated the health crisis, they also had significant economic and social impacts. The pandemic affected economies, forcing firms to adopt remote work and resulting in unemployment in sectors that relied on physical presence. Public health measures like obligatory mask-wearing and immunization campaigns significantly impacted daily habits as individuals adapted their lives based on perceived risk. But the question that I ask is how individuals react to unforeseen changes in rates of virus cases. While anticipated trends in numbers of cases might trigger similar individual responses, unexpected fluctuations in case rates might trigger spontaneous responses.

This research examines the effects of deviations in the case rates, that is, the unexplained fluctuations by vaccination rates, existing cases, and state policy, on how individuals spend time away from home in the United States. By focusing on such unexpected case rates, I will determine the impact of unforeseen changes in the trajectory of the pandemic on individual mobility behaviors, specifically the behavior of time spent away from the home. These behavioral reactions need to be known to develop policies that are more accurate in anticipating public reaction.

Literature Review

A large amount of previous research has studied the relationship between the incidence of new cases and human behavior. Previous work showed that an increase in reported cases commonly results in elevated levels of public anxiety and the associated reductions in mobility, as individuals attempt to minimize exposure risk (Bargain & Aminjonov, 2020). On the other hand, the decreasing case rate periods have overlapped with increased social activities and mobility since individuals perceive less risk (Sehra et al., 2020). This demonstrates that individuals adjust to changing behavior based on perceived risks.

The effect of government policy on behavior among individuals in reaction to COVID-19 has also been the subject of increased research. Lockdowns and other restrictions severely limited economic activity and mobility. However, depending on levels of public compliance and enforcement, these policies were differentially effective geographically (Bodas & Peleg, 2020). Political ideology, subjective economic security, and trust in the government have also, according to some research, affected compliance with public health rules and even the severity of policies (Barrios & Hochberg, 2021). Mass vaccination initiatives have also significantly helped to drive patterns of movement. The greater the number of people who are vaccinated, the more there is confidence to go out and socialize and, accordingly, a resumption of pre-pandemic behavior in the long run (Wright et al., 2021). But this is made more difficult because of vaccine hesitancy in some groups because areas with lower rates of vaccination maintained behavioral restraint.

Despite the numerous studies on the topic, few have separated the effect of unexplained fluctuations in case rates, i.e., those not explained by observable variables such as vaccination or policy interventions. The effects of overall case trends and policy compliance have been extensively studied in the literature, but little is understood about unexpected, abrupt shifts in case levels. These unanticipated changes may generate novel behavioral responses that are not captured.

The dataset generated by Chetty et al. (2023) serves as a good groundwork for this exercise using day-to-day observations on economic activity and pandemic-generated health measures, which enable effective observation of the impact of surprise changes in case rates on people's behaviors, especially time spent away from home. The goal of this research is to add new knowledge to the literature on behavioral reactions to surprise changes in pandemic patterns.

Background

The coming of the pandemic has led to an alteration of lifestyle, with authorities from across the globe imposing a series of actions to limit transmission of the disease. Added to reactions to the declining and increasing number of cases from individuals, dramatic alteration of travel behavior has ensued. Most members of the public adopted working remotely, suspending unwanted movements and social activities, while others still maintained employment activities with physical components. Also, the pandemic has

created new behaviors among consumers, including spending behaviors, an increase in online buying, and decreased utilization of public transportation.

Across states in the US, pandemic response has been varied, with lockdown, mask mandates, and vaccination campaigns policies differing. The heterogeneity offers the opportunity to study how unstable fluctuations in case rates affect individual behavior. It has also been demonstrated that political ideology affects adherence to public health interventions, with some groups defying interventions despite an increase in case numbers (Barrios & Hochberg, 2021). According to research, individuals who identify as conservative were less likely to change their behavior in response to rising case rates, while those who identify as liberal were more likely to stick to restrictions (Allcott et al., 2020).

The new case rates of the disease by date and city were a dependent variable in a regression, and vaccination levels, case levels, and policy at the state level were the independent variables. The residuals used to depict unpredictability in new case rates can be plotted as a consequence of this technique. These residuals contain the account for the unpredictable change in case rates that cannot be explained through observable reasons. Additional investigation of the effect of such residuals on home-leaving time will establish the influence of unexpected variation throughout the pandemic on weekly behavior. The results of this analysis will enable policymakers to anticipate behavioral responses to future health crises and create more effective interventions for minimizing risk while maintaining economic stability. If people react vigorously to unpredictable shifts in case numbers, then policymakers might have to enhance real-time communication tactics and issue more precise advice in times of uncertainty.

Lastly, knowing these trends will also help with economic planning, as companies can anticipate consumer behavior better in response to unexpected public health developments.

Data Source and Description

This work utilizes evidence from Chetty et al. (2023), which combines economic activity and public health data at high frequency drawn from private-sector sources. The data set provides daily data for economic indicators, COVID-19 case rates, mobility trends, as well as policy responses in various U.S. counties, cities and states. The data set

provides the ability to examine human behavioral response to the pandemic, particularly targeting mobility and time spent outside of residential places.

The significant variables of concern are:

- `New_Case_Rate_lagi,t-1`: The number of newly confirmed COVID-19 cases per 100,000 people, measured as a seven-day rolling sum in a given city.
- `New_Case_Count_lagi,t-1`: The total count of new COVID-19 cases over a seven-day period.
- `Case_Count_lagi,t-1`: The cumulative number of confirmed COVID-19 cases in a given city and week.
- `Death_Count_lagi,t-1`: The cumulative number of confirmed COVID-19 deaths in a given city and week.
- `FullVaccine_Count_lagi,t-1`: The cumulative number of fully vaccinated individuals.
- `GPS_Away_From_Homei,t`: A mobility metric indicating the time individuals spend outside their residential locations.
- Policy Variables:
 - o `schools_first_closed_lagi,t-1`: The date on which a state first mandated the closure of public K-12 schools.
 - o `nonessential_biz_first_closed_lagi,t-1`: The date on which a state first ordered the closure of all nonessential businesses.
 - o `stayathome_first_start_lagi,t-1`: The date on which a state first issued a stay-at-home order.

The data spans several weeks for several U.S. cities, enabling us to identify changes in behavior over time while holding city and time fixed. There are limitations, of course, such as biases in the mobility data (which is based on GPS tracking) and the difficulty of directly measuring individual risk perceptions.

Empirical Approach

My empirical approach consists of two regression models designed to examine the relationship between COVID-19 case rates and individual mobility behavior. The first regression model estimates the rate of new COVID-19 cases, while the second uses the error term from the first regression to explain changes in time spent away from home.

Regression Model 1: Predicting New COVID-19 Cases

I begin by estimating the determinants of new COVID-19 case rates using the following linear regression model:

$$\begin{aligned} \text{New_Case_Rate}_{t,c} &= \gamma_0 + \gamma_1 \times \text{New_Case_Count_lag}_{t-1,c} + \gamma_2 \times \text{Case_Count_lag}_{t-1,c} \\ &+ \gamma_3 \times \text{Death_Count_lag}_{t-1,c} + \gamma_4 \times \text{schools_first_closed_lag}_{t-1,c} \\ &+ \gamma_5 \times \text{nonessential_biz_first_closed_lag}_{t-1,c} \\ &+ \gamma_6 \times \text{stay_at_home_first_start_lag}_{t-1,c} + \varepsilon_{t,c} \end{aligned}$$

This regression captures how past case counts, death counts, and government interventions influence the rate of new COVID-19 cases. The inclusion of lagged variables ensures that my analysis considers past trends in predicting future outbreaks.

I see that when people start getting vaccinated, the laws aimed at reducing the spread of COVID-19 are no longer enforced. Therefore, I estimate a second model where the laws are replaced by the number of people vaccinated. The refined model is:

$$\begin{aligned} \text{New_Case_Rate}_{t,c} &= \beta_0 + \beta_1 \times \text{New_Case_Count_lag}_{t-1,c} + \beta_2 \times \text{Case_Count_lag}_{t-1,c} \\ &+ \beta_3 \times \text{Death_Count_lag}_{t-1,c} + \beta_4 \times \text{FullVaccine_Count_lag}_{t-1,c} \\ &+ \varepsilon_{t,c} \end{aligned}$$

This new specification holds the impact of vaccination rates constant on new cases while adjusting for key predictors. The error term of this model captures the remaining variation in new case rate not explained by observable factors. I use this residual in the next regression to investigate its impact on mobility.

Regression Model 2: Predicting Mobility Behavior

The second stage of my analysis examines how unpredictable fluctuations in new COVID-19 cases influence mobility behavior. I estimate the following regression for both models:

$$GPS_away_from_home_{t,c} = \delta_0 + \delta_1 \times error_ta_lag_{t-1,c} + \varepsilon_{t,c}$$

where $error_ta_lag_{t-1,c}$ is the residual from Model 1a, capturing the portion of new case rate changes that are unpredictable. The coefficient measures how unexpected case rate variations influence mobility behavior.

Similarly, for the revised Model 1b, I estimate:

$$GPS_away_from_home_{t,c} = \delta_0 + \delta_1 \times error_tb_lag_{t-1,c} + \varepsilon_{t,c}$$

Where $error_tb_lag_{t-1,c}$ is the error term from Model 1b.

This second model regressions allows us to determine if unexpected changes in case rates cause deviations in mobility patterns.

Regression Results and Interpretation

Results from Model 1a and Model 1b

Table 1: Estimating the New Case Rate

	New Case Rate Model 1A	New Case Rate Model 1B
(Intercept)	179.57380***	213.67133***
	(3.64204)	(5.10588)
New_Case_Count_lag	0.01782***	0.01747***
	(0.00032)	(0.00037)
Case_Count_lag	-0.00001	-0.00013***
	(0.00002)	(0.00003)
Death_Count_lag	-0.00943***	-0.01049***
	(0.00104)	(0.00144)
schools_first_closed_lag1	-166.77408***	
	(27.37867)	
nonessential_biz_first_closed_lag1	-54.65714+	
	(31.54411)	
stayathome_first_start_lag1	-107.40181***	
	(24.35924)	
FullVaccine_Count_lag		0.00004***
		(0.00001)
R ²	0.308	0.310
Adjusted R ²	0.307	0.310
N	7190	4982

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

The findings present deep contrasts between the first and second periods of the pandemic. During the first period, reflected in Model 1A, aggressive policy measures

curbed the rate of new cases quite effectively. During the second period (Model 1B), the policies were no longer in effect, and vaccination levels were a primary mover of case trends. Interestingly, while in Model 1B all variables are significant at the 1% level, that is not the case with Model 1A. Specifically, the variable for lagged case count loses its significance at the 10% level, and the variable for nonessential business closure is only significant at 10%. In addition, Model 1B has a larger Adjusted R^2 , suggesting that in this latter time frame the independent variables explain more of the variation in rates of new cases.

A closer look at the results reveals strong observation. The observation shows that when new cases rose in week $t-1$, they continued to rise in week t , which illustrates the transmission momentum. On the other hand, increase in death rate or deployment of public health interventions in the previous week correlated with decrease in new case rates in the following week validates the interventions. But one interesting finding in Model 1B is that an increase in the number of people vaccinated in week $t-1$ is associated with a new case rate higher than one week later in week t .

Initial impression may make one wonder about this result since vaccinations are expected to reduce case numbers. But some plausible explanations are there. An explanation is risk compensation: because there are more people vaccinated, they may feel less exposed to the virus and therefore become less cautious in their behavior, leading to more social contacts and a temporary increase in new infections. Another reason is that initial vaccination campaigns focused on high-risk populations (e.g., health care professionals and elderly persons) who were already at increased risk of being exposed to the virus. Therefore, the initial vaccine uptick might have coincided with higher levels of cases among the respective groups before the widespread immunity effects started to take their effect.

These findings reflect the shifting pandemic control dynamics. While policy constraints played an active role in the early phase, the second phase saw a shift where vaccine strategies became the most important way of curbing the spread of the disease.

Results from Model 2a and Model 2b

Table 2: Estimating the time spent away from home

	 GPS Away From Home Model 2A	 GPS Away From Home Model 2B
(Intercept)	-0.67979***	-0.53757***
	(0.00487)	(0.00433)
error_ta_lag	-0.00008***	
	(0.00002)	
error_tb_lag		-0.00022***
		(0.00001)
R ²	0.003	0.042
Adjusted R ²	0.003	0.042
N	7137	4930
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		

The negative coefficients indicate that unexpected increases in new case rates lead to decreased mobility. The coefficients are highly significant at a 1% significance level, meaning that these unexpected variations do have a high effect on the time spent away from home. I can say that the behavior of individuals differs when there are unexpected fluctuations.

Justification and Implications

The most important contribution of this study is its ability to disentangle the effect of surprise movements in COVID-19 case rates on mobility behavior. Unlike traditional models that rely on raw case numbers or policy interventions, my two-step regression model allows us to disentangle anticipated from unanticipated movements in case rates and thus provide a better picture of pandemic behavioral responses. The findings suggest that individuals do not merely react to high volumes of cases but to departures from anticipated trends. This suggests that unexpected health hazards play a critical role in determining mobility patterns.

These findings carry significant implications for public health policy. Traditional policy measures, including lockdown and mobility restrictions, were very influential in the initial stages of the pandemic, based on my regression analysis. However, as vaccine

coverage increased and governments relaxed their grip on restrictions, unexpected surges of cases also became a very good predictor of behavioral adjustments. Policy-makers must therefore, in addition to imposing policies, manage public opinion and uncertainty-reduction by good communication strategies. By providing transparent, live updates on case trends and potential threats, governments can encourage voluntary alterations in behavior without the need for strict mandates.

Research Questions for Future Studies

The findings of this study leave us open to numerous relevant research questions to which understanding the behavioral reactions to pandemics and other public health emergencies would greatly contribute. Some of these central issues must be examined through further studies:

1. How do long-term behavioral adaptations persist post-pandemic?

Initial reactions to COVID-19 outbreaks were strong, but over time, as the pandemic dragged on, people may have gotten used to the virus around them and so would respond less behaviorally to subsequent waves. Subsequent research could test whether mobility responses diminish over multiple waves of the pandemic, subjecting the concept of "risk normalization" or "pandemic fatigue" to test.

2. How do economic factors interact with public health reactions?

While my model is focused on COVID-19 case rates and policy interventions, economic stability is another important mobility behavior driver. Employment status, consumer spending habits, and financial insecurity should be studied in future work to understand their impact on people's choices to reduce mobility when cases are on the rise. A study with both health and economic considerations might provide a better understanding of crisis-induced behavioral shifts.

3. How does mobility response to sudden increases in surprise cases vary according to socioeconomic status?

My evidence suggests individuals adjust their conduct to undeserved changes in rates of COVID-19 cases, but the robustness of the adjustment may vary by population and economic group. The proposed study could examine if lower-income individuals, who

may have fewer options to work at home, respond less strongly to case increases than higher-income individuals with more employment flexibility.

4. Does media coverage determine behavioral response to case surges?

Information dissemination is an important determinant of public health compliance. Future studies need to investigate whether increased media coverage of case spikes increases behavioral responses or whether misinformation erodes compliance with public health guidelines. Understanding how various sources of information affect behavior can help develop stronger communications during health crises.

5. What is the relative effectiveness of voluntary behavioral changes versus government mandates?

Although policy interventions such as lockdowns and closures of businesses were able to reduce case numbers early in the pandemic, voluntary changes in behavior may be increasingly important in later stages. Future research can compare government-imposed versus self-reported change as a function of public perceived risk and provide valuable lessons regarding how policy interventions must evolve over time.

Conclusion

This paper examines the relationship between rates of COVID-19 cases, policy interventions, and individual mobility behavior using evidence from Chetty et al. (2023). I estimate two regression models and find that early in the pandemic, policy interventions such as school closures, business closures, and stay-at-home orders made substantial contributions towards lowering new case rates. But as vaccination campaigns continued and restrictions were lifted, such regulations became redundant with vaccination levels being the primary driving factor for case variations.

My second model of regression provides evidence that individuals respond to unexpected increases in COVID-19 cases by adjusting mobility behavior. Specifically, when rates of cases unexpectedly increase, individuals reduce time away from home, suggesting a reaction based on surprise rather than absolute case counts. The significance of these findings reinforces the impact of public opinion and risk perception upon mobility decisions under a health emergency.

In the future, policy decision-makers can make their public health communications more successful through reducing uncertainty and providing truthful current information regarding developments in the course of the pandemic. There could be more mechanisms for this type of work being followed by a study employing various other variables pertinent to consumers and employers reacting to public health interference, thus continuing to construct higher-level analysis on furthering such knowledge.

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